PADHAKU
Mathematics · Class 10 Mathematics



CHAPTER 14

Probability

NCERT Exercise Solutions

Exercise 14.1	29 Q

✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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Exercise 14.1

1 (i) Probability of an event E + Probability of the event 'not E' = ____.

✔ ANSWER

1

P(E) + P(not E) = 1 (complementary events).

1 (ii) The probability of an event that cannot happen is ____. Such an event is called ____.

✔ ANSWER

0; such an event is called an **impossible event**.

1 (iii) The probability of an event that is certain to happen is ____. Such an event is called ____.

✔ ANSWER

1; such an event is called a **sure (certain) event**.

1 (iv) The sum of the probabilities of all the elementary events of an experiment is ____.

✔ ANSWER

1

1 (v) The probability of an event is greater than or equal to ____ and less than or equal to ____.

✔ ANSWER

0 and **1**, i.e. $0 \leq P(E) \leq 1$.

2 Which of the following experiments have equally likely outcomes? Explain.

(i) A driver attempts to start a car. The car starts or does not start.

(ii) A player attempts to shoot a basketball. She/he shoots or misses the shot.

(iii) A trial is made to answer a true-false question. The answer is right or wrong.

(iv) A baby is born. It is a boy or a girl.

✔ ANSWER

(i) **Not equally likely.** A car normally starts; 'does not start' happens only when something is wrong (fault, no fuel), so the two outcomes have different chances.

(ii) **Not equally likely.** The result depends on the player's skill and practice, so 'shoots' and 'misses' are not equally probable.

(iii) **Equally likely.** The answer is chosen without any reason to favour right or wrong, so both outcomes have the same chance.

(iv) **Equally likely.** A newborn baby is a boy or a girl with the same chance.

So experiments **(iii) and (iv)** have equally likely outcomes.

3 **Why is tossing a coin considered to be a fair way of deciding which team should get the ball at the beginning of a football game?**

✔ ANSWER

When a coin is tossed, the two outcomes, head and tail, are **equally likely** (each has probability ½).

The result of a toss cannot be predicted or controlled by anyone, so neither team gets an advantage. Hence it is a fair way of deciding.

4 **Which of the following cannot be the probability of an event? (A) 2/3 (B) –1.5 (C) 15% (D) 0.7**

✔ ANSWER

(B) –1.5

The probability of an event always lies between 0 and 1 ($0 \le P(E) \le 1$). –1.5 is negative, so it cannot be a probability. (2/3, 15% = 0.15 and 0.7 all lie between 0 and 1.)

5 **If P(E) = 0.05, what is the probability of 'not E'?**

✔ ANSWER

P(not E) = 1 − P(E) = 1 − 0.05 = **0.95**

6 **A bag contains lemon flavoured candies only. Malini takes out one candy without looking into the bag. What is the probability that she takes out**

(i) an orange flavoured candy? (ii) a lemon flavoured candy?

✔ ANSWER

(i) There is no orange flavoured candy in the bag, so this is an impossible event. P(orange candy) = **0**

(ii) Every candy is lemon flavoured, so this is a sure event. P(lemon candy) = **1**

7 **It is given that in a group of 3 students, the probability of 2 students not having the same birthday is 0.992. What is the probability that the 2 students have the same birthday?**

✔ ANSWER

'Same birthday' and 'not same birthday' are complementary events.

P(same birthday) = 1 − P(not same birthday) = 1 − 0.992 = **0.008**

8 **A bag contains 3 red balls and 5 black balls. A ball is drawn at random from the bag. What is the probability that the ball drawn is (i) red? (ii) not red?**

✔ ANSWER

Total balls = 3 + 5 = 8, all equally likely.

(i) P(red) = 3/8 = **3/8**

(ii) P(not red) = 1 − 3/8 = **5/8** (there are 5 black balls, so 5/8 directly too).

9 **A box contains 5 red marbles, 8 white marbles and 4 green marbles. One marble is taken out of the box at random. What is the probability that the marble taken out will be (i) red? (ii) white? (iii) not green?**

✔ ANSWER

Total marbles = 5 + 8 + 4 = 17.

(i) P(red) = **5/17**

(ii) P(white) = **8/17**

(iii) P(green) = 4/17, so P(not green) = 1 − 4/17 = **13/17**

10 **A piggy bank contains hundred 50p coins, fifty ₹1 coins, twenty ₹2 coins and ten ₹5 coins. If it is equally likely that one of the coins will fall out when the bank is turned upside down, what is the probability that the coin (i) will be a 50 p coin? (ii) will not be a ₹5 coin?**

✔ ANSWER

Total coins = 100 + 50 + 20 + 10 = 180.

(i) P(50 p coin) = 100/180 = **5/9**

(ii) P(₹5 coin) = 10/180 = 1/18, so P(not a ₹5 coin) = 1 − 1/18 = **17/18**

11 **Gopi buys a fish from a shop for his aquarium. The shopkeeper takes out one fish at random from a tank containing 5 male fish and 8 female fish. What is the probability that the fish taken out is a male fish?**

✔ ANSWER

Total fish = 5 + 8 = 13.

P(male fish) = 5/13 = **5/13**

12 **A game of chance consists of spinning an arrow which comes to rest pointing at one of the numbers 1, 2, 3, 4, 5, 6, 7, 8, and these are equally likely outcomes. What is the probability that it will point at (i) 8? (ii) an odd number? (iii) a number greater than 2? (iv) a number less than 9?**

✔ ANSWER

Total outcomes = 8.

(i) P(8) = **1/8**

(ii) Odd numbers: 1, 3, 5, 7 → 4 outcomes. P(odd) = 4/8 = **1/2**

(iii) Numbers greater than 2: 3, 4, 5, 6, 7, 8 → 6 outcomes. P = 6/8 = **3/4**

(iv) All 8 numbers are less than 9. P = 8/8 = **1**

13 **A die is thrown once. Find the probability of getting (i) a prime number; (ii) a number lying between 2 and 6; (iii) an odd number.**

✔ ANSWER

Total outcomes = 6 (1, 2, 3, 4, 5, 6).

(i) Prime numbers: 2, 3, 5 → 3 outcomes. P = 3/6 = **1/2**

(ii) Numbers between 2 and 6: 3, 4, 5 → 3 outcomes. P = 3/6 = **1/2**

(iii) Odd numbers: 1, 3, 5 → 3 outcomes. P = 3/6 = **1/2**

14 **One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting (i) a king of red colour (ii) a face card (iii) a red face card (iv) the jack of hearts (v) a spade (vi) the queen of diamonds**

✔ ANSWER

Total outcomes = 52.

(i) Red kings: king of hearts, king of diamonds → 2. P = 2/52 = **1/26**

(ii) Face cards (jack, queen, king) = 3 × 4 = 12. P = 12/52 = **3/13**

(iii) Red face cards = 3 × 2 = 6. P = 6/52 = **3/26**

(iv) Jack of hearts → 1. P = **1/52**

(v) Spades = 13. P = 13/52 = **1/4**

(vi) Queen of diamonds → 1. P = **1/52**

15 **Five cards—the ten, jack, queen, king and ace of diamonds, are well-shuffled with their face downwards. One card is then picked up at random.**

(i) What is the probability that the card is the queen?

(ii) If the queen is drawn and put aside, what is the probability that the second card picked up is (a) an ace? (b) a queen?

✔ ANSWER

(i) Total cards = 5, one queen. P(queen) = **1/5**

(ii) After the queen is put aside, 4 cards remain: ten, jack, king, ace.

(a) P(ace) = **1/4**

(b) No queen is left, so P(queen) = **0**

16 **12 defective pens are accidentally mixed with 132 good ones. One pen is taken out at random from this lot. Determine the probability that the pen taken out is a good one.**

✔ ANSWER

Total pens = 12 + 132 = 144.

P(good pen) = 132/144 = **11/12**

17 **(i) A lot of 20 bulbs contain 4 defective ones. One bulb is drawn at random from the lot. What is the probability that this bulb is defective?**

(ii) Suppose the bulb drawn in (i) is not defective and is not replaced. Now one bulb is drawn at random from the rest. What is the probability that this bulb is not defective?

✔ ANSWER

(i) P(defective) = 4/20 = **1/5**

(ii) Now 19 bulbs remain, of which 4 are defective and 19 − 4 = 15 are not defective.

P(not defective) = **15/19**

18 **A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears (i) a two-digit number (ii) a perfect square number (iii) a number divisible by 5.**

✔ ANSWER

Total outcomes = 90.

(i) Two-digit numbers from 10 to 90 = 90 − 9 = 81. P = 81/90 = **9/10**

(ii) Perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81 → 9. P = 9/90 = **1/10**

(iii) Divisible by 5: 5, 10, 15, ..., 90 → 90 ÷ 5 = 18. P = 18/90 = **1/5**

19 **A child has a die whose six faces show the letters A, B, C, D, E, A. The die is thrown once. What is the probability of getting (i) A? (ii) D?**

✔ ANSWER

Total outcomes = 6.

(i) A appears on 2 faces. P(A) = 2/6 = **1/3**

(ii) D appears on 1 face. P(D) = **1/6**

20* **Suppose you drop a die at random on the rectangular region 3 m × 2 m (Fig. 14.6). What is the probability that it will land inside the circle with diameter 1 m? (* Not from the examination point of view.)**

✔ ANSWER

Area of rectangle = 3 × 2 = 6 m²

Radius of circle = ½ m, so area of circle = π × (½)² = π/4 m²

P(die lands inside the circle) = Area of circle ÷ Area of rectangle = (π/4) ÷ 6 = **π/24**

21 **A lot consists of 144 ball pens of which 20 are defective and the others are good. Nuri will buy a pen if it is good, but will not buy if it is defective. The shopkeeper draws one pen at random and gives it to her. What is the probability that (i) She will buy it? (ii) She will not buy it?**

✔ ANSWER

Total pens = 144; good pens = 144 − 20 = 124.

(i) P(she buys) = P(good) = 124/144 = **31/36**

(ii) P(she does not buy) = P(defective) = 20/144 = **5/36** (also 1 − 31/36 = 5/36)

22 **Refer to Example 13 (two dice thrown together, 36 equally likely outcomes).**

(i) Complete the table of probabilities for the event 'Sum on 2 dice' = 2, 3, 4, ..., 12.

(ii) A student argues that 'there are 11 possible outcomes 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 and 12. Therefore, each of them has a probability 1/11.' Do you agree with this argument? Justify your answer.

✔ ANSWER

(i) Total outcomes = 36. Count the pairs giving each sum (e.g. sum 4: (1,3), (2,2), (3,1) → 3 ways).

Sum on 2 dice	2	3	4	5	6	7	8	9	10	11	12
Probability	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

(ii) **No.** The eleven sums are **not equally likely**. For example, sum 2 comes only from (1,1), 1 way, while sum 7 comes from (1,6), (2,5), (3,4), (4,3), (5,2), (6,1), 6 ways. So each sum does not have probability 1/11; the probabilities are as in the table above.

23 **A game consists of tossing a one rupee coin 3 times and noting its outcome each time. Hanif wins if all the tosses give the same result i.e., three heads or three tails, and loses otherwise. Calculate the probability that Hanif will lose the game.**

✔ ANSWER

Possible outcomes: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT → 8 outcomes, all equally likely.

Hanif wins for HHH or TTT → 2 outcomes. P(wins) = 2/8 = 1/4

P(Hanif loses) = 1 − 1/4 = **3/4** (6 of the 8 outcomes)

24 **A die is thrown twice. What is the probability that (i) 5 will not come up either time? (ii) 5 will come up at least once? [Hint: Throwing a die twice and throwing two dice simultaneously are treated as the same experiment]**

✔ ANSWER

Total outcomes = 6 × 6 = 36.

(i) For 5 not to come up either time, each throw has 5 choices (1, 2, 3, 4, 6): 5 × 5 = 25 outcomes.

P(5 does not come up either time) = **25/36**

(ii) P(5 comes up at least once) = 1 − 25/36 = **11/36**

25 **Which of the following arguments are correct and which are not correct? Give reasons for your answer.**

(i) If two coins are tossed simultaneously there are three possible outcomes—two heads, two tails or one of each. Therefore, for each of these outcomes, the probability is 1/3.

(ii) If a die is thrown, there are two possible outcomes—an odd number or an even number. Therefore, the probability of getting an odd number is 1/2.

✔ ANSWER

(i) **Incorrect.** The outcomes can be grouped like this, but the three groups are not equally likely. 'One of each' can happen in two ways, HT or TH, while 'two heads' (HH) and 'two tails' (TT) happen in one way each. The four outcomes HH, HT, TH, TT are equally likely, so P(two heads) = 1/4, P(two tails) = 1/4, P(one of each) = 2/4 = 1/2.

(ii) **Correct.** Odd numbers (1, 3, 5) and even numbers (2, 4, 6) are 3 each out of 6 equally likely outcomes, so the two outcomes 'odd' and 'even' are equally likely and P(odd) = 3/6 = 1/2.

🎉 **Chapter 14 done! Revise the answers once more tomorrow.**